Arithmetic Sequences and Series

Section 1: General Term of an Arithmetic Sequence

1 State the formula for the n-th term a_n of an arithmetic sequence with first term a_1 and common difference d.

a. $a_n = a_1 + (n-1)d$

b. $a_n = a_1 \cdot d^{n-1}$

c. $a_n = a_1 + nd$

d. $a_n = \frac{a_1 + a_n}{2}$

2 Find the 20th term of the arithmetic sequence $4, 7, 10, 13, \ldots$

a. 61

b. 58

c. 64

d. 60

3 In an arithmetic sequence, $a_3 = 11$ and $a_8 = 31$. Find the first term a_1 and common difference d.

a. $a_1 = 3, d = 4$

b. $a_1 = 4, d = 3$

c. $a_1 = 2, d = 5$

d. $a_1 = 1, d = 4$

Section 2: Sum of an Arithmetic Series

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4 State the two standard formulas for the sum S_n of the first n terms of an arithmetic series.

a. $S_n = \frac{n}{2}(a_1 + a_n)$ and $S_n = \frac{n}{2}[2a_1 + (n-1)d]$

b. $S_n = \frac{a_1(1-r^n)}{1-r}$

c. $S_n = n(a_1 + a_n)$

d. $S_n = \frac{a_1 + a_n}{2}$

5 Find the sum of the first 30 terms of the arithmetic series $2 + 5 + 8 + 11 + \ldots$

a. 1365

b. 1350

c. 2730

d. 1410

6 Find the sum of the positive integers from 1 to 100.

a. 5050

b. 5000

c. 5100

d. 10100

Section 3: Sigma Notation and Number of Terms

7 Evaluate $\sum_{k=1}^{25} (4k - 1)$.

a. 1275

b. 1300

c. 1250

d. 2550

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8 How many terms are in the finite arithmetic sequence $7, 11, 15, \ldots, 87$?

a. 21

b. 20

c. 22

d. 19

9 Evaluate $\sum_{k=5}^{20} 3k$.

a. 600

b. 570

c. 630

d. 300

Section 4: Word Problems and Real-World Applications

10 An auditorium has 20 rows of seats. The first row has 15 seats, and each subsequent row has 2 more seats than the row before it. How many total seats are in the auditorium?

a. 680

b. 660

c. 700

d. 580

11 A runner trains by running 3 miles on day 1, and increases her daily distance by 0.5 miles each day. How many total miles will she have run in 14 days?

a. 87.5 miles

b. 91 miles

c. 84 miles

d. 95.5 miles

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12 A clock chimes the number of hours on the hour (1 chime at 1:00, 2 chimes at 2:00, ..., 12 chimes at 12:00). How many total chimes does it make in a 12-hour period?

a. 78

b. 72

c. 84

d. 66

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